

Assignment #4

Studies Done:

1. Chi-Square difference to test whether the two-factor model of GRIT fit better than the single factor model of GRIT.
2. Correlational test between GRIT-S and the Big Five traits
3. Ordinal logistic regression predicting education attainment from GRIT-S (age as covariate)

1. Chi Square difference

Original

“A confirmatory factor analysis supported the two-factor model of grit. The two subscales, Consistency of Interest and Perseverance of Effort, were first-order latent factors that loaded on a second-order latent factor called Grit. We compared this two-factor model to a more parsimonious model in which all eight items loaded on a single latent factor. Structural equation models were run with AMOS Version 6.0 (Arbuckle, 2005) using the maximum-likelihood method. **The two-factor model, $\chi^2(19, N = 1,554) = 188.52, p < .001$, fit the data significantly better than did the single-factor model, $\chi^2(20, N = 1,554) = 380.45, p < .001$ as indicated by a significant chi-square difference, $\chi^2(1) = 191.93, p < .001$.** Other fit indexes suggest a good fit for the two-factor model, RMSEA = .076 (90% CI = .066–.086), CFI = .96.”(Duckworth and Quinn 2009).

My Thoughts?

The authors conducted a chi square test to compare a two-factor hierarchical model of GRIT with a more parsimonious single-factor model. The first model separated the concepts of GRIT into 2 subsets: perseverance of effort and consistency of interest, whereas the single factor model had these 2 subsets loaded into a single factor in the model. Their Chi square analysis showed that relaxing the constraints reduced the model misfit when compared to the single factor specification. However, the authors wording that the two-factor model fit the data “significant better” is misleading given that chi-squares are extremely sensitive to large sample sizes and there was a large sample size used for this test. As a result the statistical significance here reflects differences in the model misfit rather than the fact that the two-factor model being a more accurate representation of the underlying construct.

Rewritten:

“A confirmatory factor analysis between the two factor and single factor model of GRIT was statistically clear indicating a reduced model misfit when using the 2-factor model. The two subscales, Consistency of Interest and Perseverance of Effort, were first order latent factors that loaded on a second-order latent factor called Grit. We compared this two-factor model to a more parsimonious model in which all eight items loaded on a single latent factor. Structural equation models were run with AMOS Version 6.0 (Arbuckle, 2005) using the maximum-likelihood method. **The chi square difference between the two-factor model $\chi^2(19, N = 1,554) = 188.52, p < .001$ and the single-factor model $\chi^2(20, N = 1,554) = 380.45, p < .001$, show a clear sign of statistical significance $\chi^2(1) = 191.93, p < .001$ indicating that the two factor model showed reduced misfit.** Other fit indexes suggest a good fit for the two-factor model, RMSEA = .076 (90% CI = .066–.086), CFI = .96.”(Duckworth and Quinn 2009).

2. Correlational test between GRIT-S and Big Five

Original

“As predicted, the Grit–S correlated more with BFI Conscientiousness ($r = .77, p < .001$) than with Neuroticism ($r = -.40, p < .001$), Extraversion ($r = .20, p < .001$), Agreeableness ($r = .24, p < .001$), or Openness to Experience ($r = .06, p = .03$).”(Duckworth and Quinn 2009).

My Thoughts?

The authors report a series of zero-order correlations describing associations between GRIT-S and the Big Five traits. The statistics are reported accurately and do not imply causality or predictive relationships. I wonder if the usage of the terminology “As predicted...” could imply that this offers some sort of support for a pre-existing theory which is frowned upon to do using correlations and p values. However, I think the authors were simply acknowledging that this relationship is what they predicted.

Moreover, I am not sure if a correlational study can have confidence intervals displayed, which I noticed was not present for this statistical reporting.

Rewritten

Association between GRIT-S and Conscientiousness was statistically clear with strong correlation coefficient ($r = .77, p < .001$). Furthermore, there was a statistically clear moderate correlations with Neuroticism ($r = -.40, p < .001$), Extraversion ($r = .20, p < .001$), Agreeableness ($r = .24, p < .001$), or Openness to Experience ($r = .06, p = .03$).

3. Ordinal Logistic Regression Predicting Education Attainment from GRIT-S

“In an ordinal logistic regression predicting educational attainment from Grit–S scores and using age as a covariate, both Grit–S ($B = 0.21$, odds ratio [OR] = 1.23, $p < .001$) and age ($B = 0.22$, OR = 1.25, $p < .001$) were significant predictors. That is, participants who scored 1 SD higher in grit than same-aged peers were 23% more likely to have attained more education.”(Duckworth and Quinn 2009)

My Thoughts?

The ordinal logistic regression tells you the odds of being in higher category of educational attainment with each increase in GRIT while the other variables are held constant. After ensuring age and other big five personality traits are controlled for, higher grit scores are associated positively with higher education attainment categories. We cannot make any assumptions that GRIT increases higher educational attainment. I don't see any problem with the way these statistics have been presented. However, I do think the wording might hint at a stronger meaning than what exists. Stating that GRIT can predict higher educational attainment seems to be a strong claim to make from a. association. Thus, I do believe this presentation needs to be written to match a more defensible statement.

Rewritten

In an ordinal logistic regression, higher GRIT-S scores were associated with greater likelihoods of being in a higher educational attainment category after controlling for age ($B = 0.21$, odds ratio [OR] = 1.23, $p < .001$) and age ($B = 0.22$, OR = 1.25, $p < .001$). This showcases a positive association between GRIT and educational attainment, but we cannot say with certainty that GRIT predicts educational outcomes.

Reference:

Duckworth AL, Quinn PD. 2009. Development and Validation of the Short Grit Scale (Grit–S). *Journal of Personality Assessment*. 91(2):166–174.
<https://doi.org/10.1080/00223890802634290>